

class test - 01

11/06/26

① computer vision is a field of ~~artificial~~ intelligence that allows computers to capture process - Analyze, and understand images, to make decisions.

a> Detect and recognize objects.
understand scenery of images content

b> face recognition
self-driving cars.

image acquisition
feature extraction
object detection

a> feature extraction is the process of the identifying important char from the image.

b> object recognition identifies and classifies object in image.

a> Low-level processing improves image quality through operations like removal noise, filtering and enhancement.

b> Mid level

Extracts features and segment images

High level

interprets image and recognize

output image features

discuss with

the sn

(4)

a> A pixel (picture element) is

unit of a digital image.

b> Higher resolution means more pixels resulting in sharper and more detailed images.

(5)

a> Role of image sensor, is the conversion of incoming light into electrical signals.

b> Two image acquisition devices

* Digital camera

* Webcam

(6)

a> Sampling converts a continuous image into a grid of pixels.

b> Quantization assigns discrete intensity values to pixels.

a) perspective projection maps a 3D scene onto a 2D image while preserving depth perception.

b) Illumination provides the light needed to capture visible objects details.

8. a) image processing is the manipulation of images to improve their quality or extract information.

b) image processing or modifying enhance images

computer vision understands and interprets image.

Output is an improved image

outputs information to make decision

a) computer vision analyzes medical images (X-rays, CT scans, MRI) to detect diseases.

67 It detects roads, traffic signals and
to monitor over the vehicle to safe
driving.

- (10) a) Because every digital image is
made up pixels, each storing brightness
(or) colour information.
- b) Sampling determines the no. of pixels
captured.
- * resolution determines the no. of pixels
accuracy.
 - * High sampling and resolution improve
object detection and recognition.